PROPER BASE CHANGE FOR UNBOUNDED COMPLEXES OF Z/NZ-MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to a quasi-compact base with the dimension bound, the choice of the K-injective representative, the degreewise vanishing of the higher pushforwards of the base-changed terms, and the resulting identification of complexes.

Editorial note. The label lemma-proper-base-change-mod-n names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the displayed base change equation of the chapter and is delivered verbatim; the six cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Finiteness of the cohomological dimension of a proper morphism. Same chapter, the lemma bounding the cohomological dimension of a proper morphism through the fibre dimension, together with Morphisms, the boundedness of the fibre dimension of a morphism of finite type. Supplies the finite cohomological dimension of f_* and f'_* in the sense needed for unbounded derived categories.
- K-injective resolutions with injective terms. Injectives, the theorem providing a K-injective complex of injective modules representing a given complex. Supplies the representative I^bullet of E used in the computation.
- Computation of an unbounded right derived functor on a complex of acyclics, and the usual proper base change theorem. Derived Categories, the lemma allowing the derived functor to be computed on a complex of acyclics, and Same chapter, the proper base change theorem. Supplies the reduction of the computation to f'_*(g')^{-1}I^bullet and the identification of that complex with g^{-1}f_*I^bullet.

Source reference. Original lemma 07K7: sources/stacks-chow/tags.txt, source lines 9647–9647. Prerequisite pointer invoked at proof line(s) 18919; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 07HF: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18332–18332. Prerequisite pointer invoked at proof

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line(s) 18893; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 079P: sources/stacks-chow/tags.txt, source lines 9279–9279. Prerequisite pointer invoked at proof line(s) 18925; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 095U: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18786–18786. Prerequisite pointer invoked at proof line(s) 18904; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A3V: sources/stacks-chow/tags.txt, source lines 12753–12753. Prerequisite pointer invoked at proof line(s) 18901; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 095T: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18610–18610. Prerequisite pointer invoked at proof line(s) 18928; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of schemes. Let $g : Y' \to Y$ be a morphism of schemes. Set $X' = Y' \times_Y X$ and denote $f' : X' \to Y'$ and $g' : X' \to X$ the projections. Let $n \geq 1$ be an integer. Let $E \in D(X_{étale}, \mathbf{Z}/n\mathbf{Z})$. Then the base change map (07HF) $g^{-1}Rf_*E \to Rf'_*(g')^{-1}E$ is an isomorphism.*

Proof. It is enough to prove this when Y and Y' are quasi-compact. By Morphisms, Lemma 0A3V we see that the dimension of the fibres of $f : X \to Y$ and $f' : X' \to Y'$ are bounded. Thus Lemma 095U implies that

$$f_* : Mod(X_{étale}, \mathbf{Z}/n\mathbf{Z}) \longrightarrow Mod(Y_{étale}, \mathbf{Z}/n\mathbf{Z})$$

and

$$f'_* : Mod(X'_{étale}, \mathbf{Z}/n\mathbf{Z}) \longrightarrow Mod(Y'_{étale}, \mathbf{Z}/n\mathbf{Z})$$

have finite cohomological dimension in the sense of Derived Categories, Lemma 07K7. Choose a K-injective complex $\mathcal{I}^\bullet$ of $\mathbf{Z}/n\mathbf{Z}$-modules each of whose terms $\mathcal{I}^n$ is an injective sheaf of $\mathbf{Z}/n\mathbf{Z}$-modules representing E. See Injectives, Theorem 079P. By the usual proper base change theorem we find that $R^q f'_*(g')^{-1}\mathcal{I}^n = 0$ for $q > 0$, see Theorem 095T. Hence we conclude by Derived Categories, Lemma 07K7 that we may compute $Rf'_*(g')^{-1}E$ by the complex $f'_*(g')^{-1}\mathcal{I}^\bullet$. Another application of the usual proper base change theorem shows that this is equal to $g^{-1}f_*\mathcal{I}^\bullet$ as desired. □